

Project report

Transcriptional response of *Drosophila* treated with Canurta formulations

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Feb 10th, 2026

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Overview

The goal of this project was to extract, prepare and sequence RNA for RNA-seq based differential gene expression analysis in *Drosophila melanogaster* in order to characterize the response to treatment with Canurta preparations.

Results

RNA isolation, RNA-seq library preparation, sequencing and analysis were conducted for four biological replicates in both male and female flies for two treatments with corresponding controls.

Statistics for the 32 samples and their RNA extraction are summarized in Table 1. The initial round of extraction on the male / CFA block of samples using the Qiagen RNeasy Plus provided relatively low yields, though these yields were still sufficient for RNA-seq library preparation. The NEB Monarch Total RNA prep was employed for all remaining samples and provided a much higher yield of RNA. Though the quality of RNA isolated from both protocols was deemed to be high (RNA integrity number, RIN range = 9.3-9.7 for Qiagen, 9.0-9.8 for NEB), due to the higher yield the NEB protocol is recommended for future work. As the block of samples prepared using the Qiagen RNeasy Plus protocol included all controls and treatments for that treatment and sex combination, the method will not be a confounder for the differential expression analysis in that treatment / sex combination.

The resulting libraries were sequenced to an average depth of 34 million read pairs (Table 2). The samples prepared using the Qiagen RNAeasy Plus protocol yielded slightly lower number of reads, averaging 23 million read pairs per replicate. Principal components analysis and heat map analysis (Figs 2 and 3) of transcript abundance per sample showed a strong grouping based on sex. The variance captured by the second principal component was unclear.

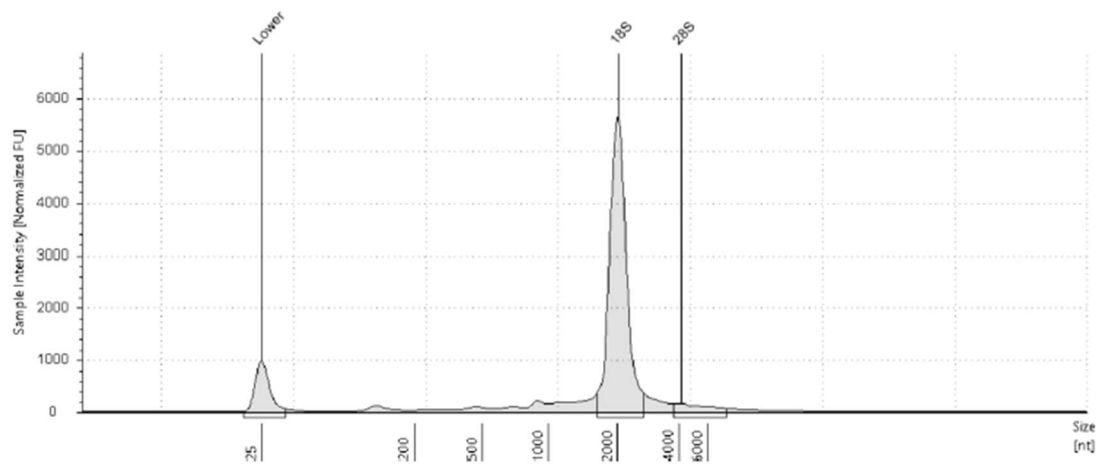
Raw data, read counts and differential expression results are available at:

https://bioinfo.nrc.ca/canurta_zebrafish

Table 1. Sample and RNA extraction statistics

Sample name	Fly heads	Extraction Date	Extraction Kit	RNA Qubit (ng/uL)	Elution vol (uL)	Total RNA yield (ng)
CFA 1mM_MC-1	30	Sept 4/ 2025	Qiagen RNeasy Plus	1.599	30	47.97
CFA 1mM_MC-2	29	Sept 4/ 2025	Qiagen RNeasy Plus	3.117	30	93.51
CFA 1mM_MC-3	30	Sept 4/ 2025	Qiagen RNeasy Plus	5.613	30	168.39
CFA 1mM_MC-4	29	Sept 4/ 2025	Qiagen RNeasy Plus	9.258	30	277.74
CFA 1mM_MT-1	27	Sept 4/ 2025	Qiagen RNeasy Plus	3.917	30	117.51
CFA 1mM_MT-2	24	Sept 4/ 2025	Qiagen RNeasy Plus	3.545	30	106.35
CFA 1mM_MT-3	28	Sept 4/ 2025	Qiagen RNeasy Plus	3.347	30	100.41
CFA 1mM_MT-4	29	Sept 4/ 2025	Qiagen RNeasy Plus	1.669	30	50.07
CFA 1mM_FC-1	30	Sept 11/2025	NEB Monarch Total RNA prep	122	50	6100
CFA 1mM_FC-2	30	Sept 11/2025	NEB Monarch Total RNA prep	82.6	50	4130
CFA 1mM_FC-3	30	Sept 11/2025	NEB Monarch Total RNA prep	116	50	5800
CFA 1mM_FC-4	30	Sept 11/2025	NEB Monarch Total RNA prep	134	50	6700
CFA 1mM_FT-1	28	Sept 11/2025	NEB Monarch Total RNA prep	85.8	50	4290
CFA 1mM_FT-2	75	Sept 11/2025	NEB Monarch Total RNA prep	79.4	50	3970
CFA 1mM_FT-3	25	Sept 11/2025	NEB Monarch Total RNA prep	77.8	50	3890
CFA 1mM_FT-4	25	Sept 11/2025	NEB Monarch Total RNA prep	70.2	50	3510
CNR-401 1mM_MC-1	28	Sept 12/2025	NEB Monarch Total RNA prep	32	50	1600
CNR-401 1mM_MC-2	28	Sept 12/2025	NEB Monarch Total RNA prep	39	50	1950
CNR-401 1mM_MC-3	28	Sept 12/2025	NEB Monarch Total RNA prep	35.8	50	1790
CNR-401 1mM_MC-4	28	Sept 12/2025	NEB Monarch Total RNA prep	33.2	50	1660
CNR-401 1mM_MT-1	26	Sept 12/2025	NEB Monarch Total RNA prep	43.6	50	2180
CNR-401 1mM_MT-2	26	Sept 12/2025	NEB Monarch Total RNA prep	25.8	50	1290
CNR-401 1mM_MT-3	26	Sept 12/2025	NEB Monarch Total RNA prep	28.4	50	1420
CNR-401 1mM_MT-4	26	Sept 12/2025	NEB Monarch Total RNA prep	37	50	1850
CNR-401 1mM_FC-1	28	Sept 12/2025	NEB Monarch Total RNA prep	90.8	50	4540
CNR-401 1mM_FC-2	28	Sept 12/2025	NEB Monarch Total RNA prep	74.2	50	3710
CNR-401 1mM_FC-3	28	Sept 12/2025	NEB Monarch Total RNA prep	64.2	50	3210
CNR-401 1mM_FC-4	28	Sept 12/2025	NEB Monarch Total RNA prep	65.2	50	3260
CNR-401 1mM_FT-1	24	Sept 12/2025	NEB Monarch Total RNA prep	56	50	2800
CNR-401 1mM_FT-2	24	Sept 12/2025	NEB Monarch Total RNA prep	49.4	50	2470
CNR-401 1mM_FT-3	24	Sept 12/2025	NEB Monarch Total RNA prep	45.8	50	2290
CNR-401 1mM_FT-4	24	Sept 12/2025	NEB Monarch Total RNA prep	65.2	50	3260

B1: CFA 1mM MC-1



A1: Canurta_Flies-1 (CFA 1mM_MC-1)

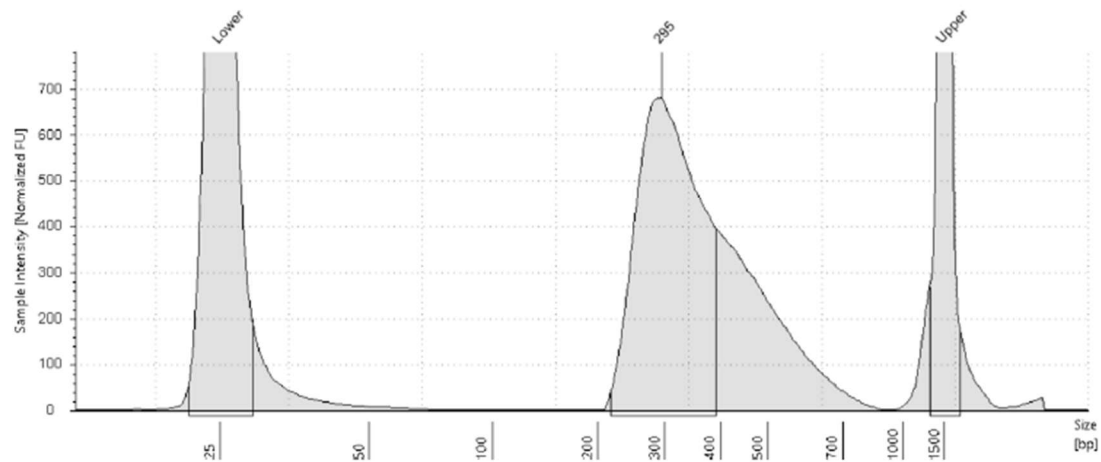


Figure 1. Tapestation-based electrophoretic quality control of input RNA and RNA-seq libraries. Upper pane: Representative input RNA sample run on a high sensitivity RNA screentape. Lower pane: Representative final RNA-seq library run on DNA 1000 screentape.

Table 2. RNA-seq library input and sequencing yield

Sample	Total input RNA (ng)	Number of read pairs
CFA_1mM_MC-1	47.97	12356335
CFA_1mM_MC-2	93.51	24898442
CFA_1mM_MC-3	168.39	18136815
CFA_1mM_MC-4	277.74	27966821
CFA_1mM_MT-1	117.51	32164773
CFA_1mM_MT-2	106.35	26878366
CFA_1mM_MT-3	100.41	28357787
CFA_1mM_MT-4	50.07	16214444
CFA_1mM_FC-1	1000	49921257
CFA_1mM_FC-2	1000	21251429
CFA_1mM_FC-3	1000	40921246
CFA_1mM_FC-4	1000	58688213
CFA_1mM_FT-1	1000	39669955
CFA_1mM_FT-2	1000	40157631
CFA_1mM_FT-3	1000	49855539
CFA_1mM_FT-4	1000	39270286
CNR-401_1mM_MC-1	800	27944947
CNR-401_1mM_MC-2	975	24890383
CNR-401_1mM_MC-3	895	32087082
CNR-401_1mM_MC-4	830	19983607
CNR-401_1mM_MT-1	1000	34421491
CNR-401_1mM_MT-2	645	26955860
CNR-401_1mM_MT-3	710	50970477
CNR-401_1mM_MT-4	925	60043860
CNR-401_1mM_FC-1	1000	30113730
CNR-401_1mM_FC-2	1000	37427059
CNR-401_1mM_FC-3	1000	23228629
CNR-401_1mM_FC-4	1000	36166426
CNR-401_1mM_FT-1	1000	33673534
CNR-401_1mM_FT-2	1000	51426098
CNR-401_1mM_FT-3	1000	48755488
CNR-401_1mM_FT-4	1000	21754540

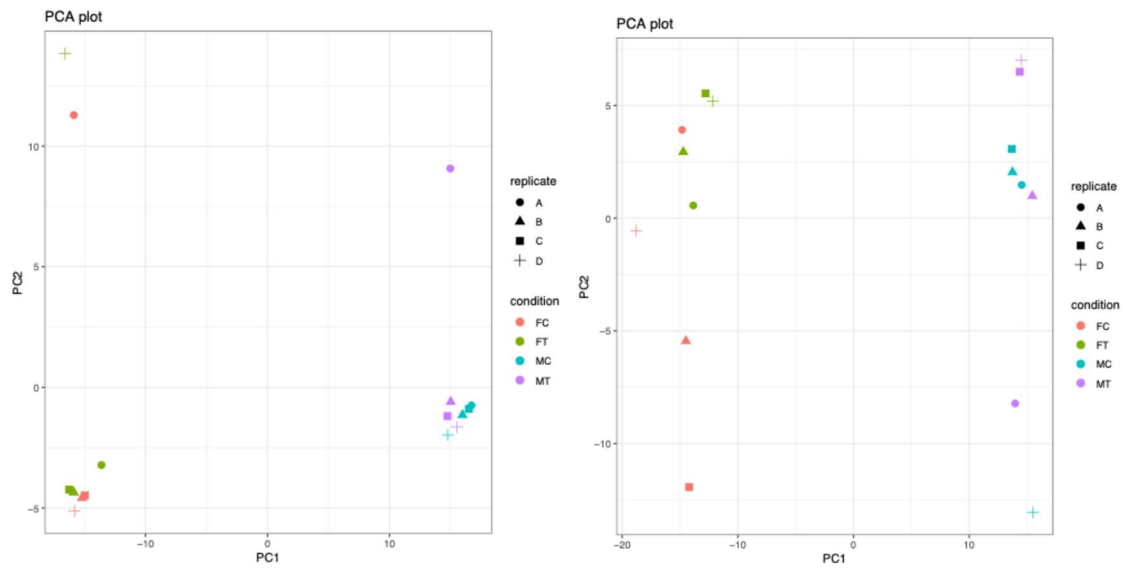


Figure 2. Principal component analysis of transcript abundance for each replicate within the CNR treatment (left) and CFA treatment (right).

Material and methods

RNA extraction and sequencing library preparation

RNA was extracted in 8 sample batches corresponding to each sex / treatment group. The first batch (Male / CFA treatment) using RNA was extracted using the Qiagen RNeasy extraction kit according to manufacturers instructions and the second using NEB Monarch Total RNA. For both extraction methods ~100uL of initial buffer (Qiagen=Buffer RLT+ B-mercaptoethanol, NEB= Protection Buffer) from each kit was added to the tubes containing the flies and the samples were homogenized for ~ 30 seconds using a small pestle. The remaining buffer volume was then added to each sample based on the protocol. The samples were briefly homogenized again for 10 seconds prior to proceeding with the remainder of the protocol according to manufacturers instructions. RNA-seq libraries were prepared using the Illumina Stranded mRNA by Ligation Kit according to manufacturer's instruction. Libraries were sequenced on an Illumina Novaseq X+ instrument using v1.5 2x150 bp chemistry.

Differential gene expression (DEG) analysis

Raw sequencing read files were trimmed using cutadapt (Martin, 2011). Fastqc (Andrews, 2010) was used to analyze the quality of the reads before and after trimming. Reads were aligned and quantified with STAR (Dobin et al., 2013; Dobin and Gingeras, 2015) against the *Drosophila_melanogaster* BDGP6.46 reference (see paths below) in quant mode. Raw read per gene counts from STAR were formatted into a matrix (gene x sample) and imported into DESeq2 (Love et al., 2014) for all differential gene expression analyses and pairwise comparisons. An adjusted p-value (p_{adj}, q) false discovery rate

(FDR) of 0.05 (Benjamini and Hochberg) was used as the cutoff value for downstream analysis.

Reference files were obtained from Ensembl:

https://oct2024.archive.ensembl.org/Drosophila_melanogaster/Info/Index

Exact paths to the reference files:

https://ftp.ensembl.org/pub/release-113/fasta/drosophila_melanogaster/dna/Drosophila_melanogaster.BDGP6.46.dna.toplevel.fa.gz

https://ftp.ensembl.org/pub/release-113/gtf/drosophila_melanogaster/Drosophila_melanogaster.BDGP6.46.113.gtf.gz

https://ftp.ensembl.org/pub/release-113/fasta/drosophila_melanogaster/cdna/Drosophila_melanogaster.BDGP6.46.cdna.all.fa.gz

Appendix 1. Abbreviations

MC-1 = Male Control group one
MC-2 = Male Control group two
MC-3 = Male Control group three
MC-4 = Male Control group four
MT-1 = Male Test group one
MT-2 = Male Test group two
MT-3 = Male Test group three
MT-4 = Male Test group four
FC-1 = Female Control group one
FC-2 = Female Control group two
FC-3 = Female Control group three
FC-4 = Female Control group four
FT-1 = Female Test group one
FT-2 = Female Test group two
FT-3 = Female Test group three

FT-4 = Female Test group four

References

Andrews, Simon. "FastQC: a quality control tool for high throughput sequence data. 2010." (2017).

Dobin, Alexander et al. "STAR: ultrafast universal RNA-seq aligner." *Bioinformatics (Oxford, England)* vol. 29,1 (2013): 15-21. doi:10.1093/bioinformatics/bts635

Dobin, Alexander, and Thomas R Gingeras. "Mapping RNA-seq Reads with STAR." *Current protocols in bioinformatics* vol. 51 11.14.1-11.14.19. 3 Sep. 2015, doi:10.1002/0471250953.bi1114s51

Love, Michael I., Wolfgang Huber, and Simon Anders. "Moderated estimation of fold change and dispersion for RNA-seq data with DESeq2." *Genome biology* 15 (2014): 1-21.

Martin, Marcel. "Cutadapt removes adapter sequences from high-throughput sequencing reads." *EMBnet. journal* 17.1 (2011): 10-12.

Wu, Tianzhi et al. "clusterProfiler 4.0: A universal enrichment tool for interpreting omics data." *Innovation (Cambridge (Mass.))* vol. 2,3 100141. 1 Jul. 2021, doi:10.1016/j.xinn.2021.100141